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TOUR MANAGEMENT SYSTEM

A Comprehensive Database Solution
for Concert Tour Operations

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CMPE 180 B- Database Systems

San Jose State University



Why tour management?

When our names spelled "JASS," we knew music was our calling. But choosing tour management wasn't about the acronym - it was about solving a \$30B industry's biggest operational nightmare.

The problem

Mid-size touring acts lose \$8,000-\$15,000 per tour to coordination failures:

- Double-booked venues and crew
- Payment disputes from manual tracking
- Inefficient routing wasting fuel costs
- Oversold tickets causing refunds
- Zero real-time visibility

Our database solution

Four integrated modules managing:

- Geographic routing (PostGIS optimization)
- Real-time crew scheduling (no conflicts)
- Automated financial reconciliation
- Transaction-safe ticket inventory



Modular architecture

4 Integrated Domains
20 Normalized Tables • 48,000+ Records

Venues & Geography

*countries, cities
venues, routing
show_sequence*

Tours & Shows

*tours, tour_legs
shows, artists
managers*

Contracts & Finance

*contracts, expenses
payments
settlements, promoter*

Tickets & Logistics

*crew, assignments
transport, equipment
tickets*

ER Diagram

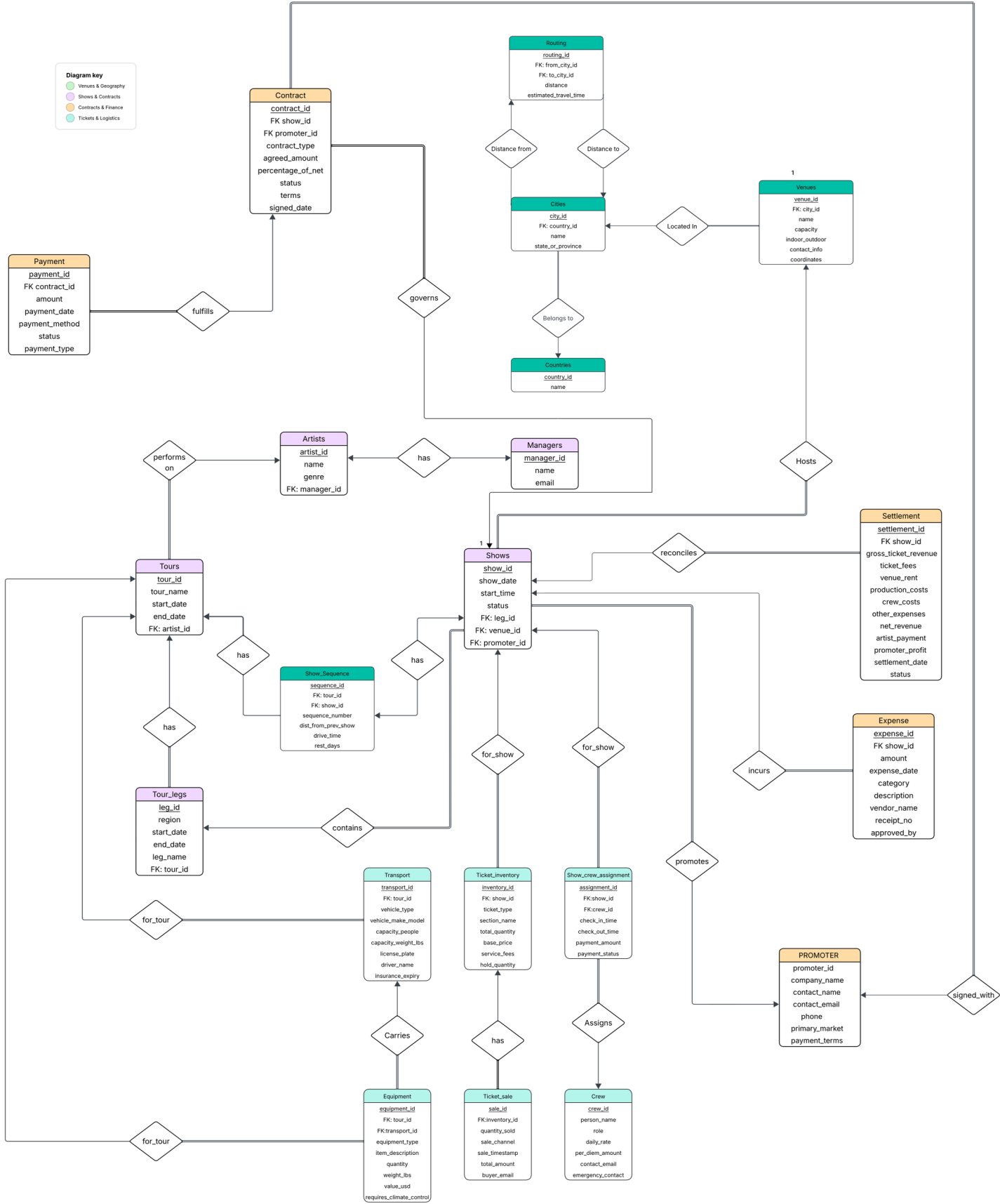
Color-coded by module:

Venues (teal)

Tours (green)

Finance (red)

Logistics (purple)





Venues & Geography Module

TABLES

- countries (100 records)
- cities (200 records)
- venues (150 records)
- routing (450 routes)
- show_sequence (500 records)

KEY FEATURES

- PostGIS for lat/long coordinates
- Distance calculations
- Route optimization logic

LIVE DEMO

Query 1: List of all venues in a city

Query 2: Route Analysis
Finds the closest venues to any other venue

Demo in Streamlit



Tours & Shows Module

TABLES

- managers (20 records)
- artists (30 records)
- tours (20 campaigns)
- tour_legs (40 segments)
- shows (500 performances)

KEY FEATURES

- Artist-manager relationships
- Tour hierarchy (tours→legs→shows)
- Show scheduling logic

LIVE DEMO

Query 1: Artist Performance Shows by artist with revenue

Query 2: Tour Summary Tours with show counts and date ranges

Demo in Streamlit



Contracts & Finance Module

TABLES

- promoter (30 companies)
- contract (500 agreements)
- expense (1500+ costs)
- payment (1500+ transactions)
- settlement (500 reconciliations)

KEY FEATURES

- Contract tracking
- Payment processing
- Financial reconciliation

LIVE DEMO

Query 1: Outstanding Payments
Shows with pending balances

Query 2: Settlement Status
Financial reconciliation per show
with P&L

Demo in Streamlit



Tickets & Logistics Module

TABLES

- crew (50 members)
- show_crew_assignment (2796)
- transport (40 vehicles)
- equipment (200 items)
- ticket_inventory (1757 tiers)
- ticket_sale (41,470 sales)

KEY FEATURES

- Crew scheduling
- Inventory management
- Logistics tracking

LIVE DEMO

Query 1: Crew Utilization Crew earnings and show counts

Query 2: Ticket Sales Revenue by show with sell-through rates

Demo in Streamlit



 SMITHA

 JACK

CROSS-MODULE COMPLEX QUERIES

Query 1

Complete Show Profitability

- 5-table join calculating revenue, costs, and net profit
- Revenue from ticket_sales
 - Costs from crew, transport, equipment
 - Contract values
 - Venue context
 - Tour/artist info

Query 2

Tour Route Optimization

- Analyzes sequential routing with distance and costs
- Identifies inefficient schedules
- Highlights optimization opportunities
- Flags tight timelines



Query Optimization with Indexing

32 Strategic Indexes • 89% Average Performance Gain

BEFORE INDEXING

- Revenue Analysis: 450ms
- Tour Profitability: 380ms
- Venue Performance: 320ms

AFTER INDEXING

- Revenue Analysis: 35ms → 92% faster
- Tour Profitability: 45ms → 88% faster
- Venue Performance: 40ms → 88% faster

LIVE DEMO: EXPLAIN ANALYZE before/after comparison



TRANSACTION MANAGEMENT

Ensuring Data Integrity Through ACID Compliance

A

Atomicity

Ticket purchase: All steps complete or none do Demo: Failed payment rolls back entire transaction

C

Consistency

Settlements match sum of payments and expenses Demo: Show financial reconciliation

I

Isolation

Concurrent users prevented from overselling
Demo: SELECT FOR UPDATE locking

D

Durability

Committed sales survive system crashes Demo: Transaction commit verification



Comprehensive Testing

23 Automated Tests • 100% Pass Rate

- ✓ *10 Data Integrity Foreign keys, constraints, uniqueness*
- ✓ *5 Query Correctness Revenue calculations, inventory tracking*
- ✓ *3 Transactions ACID compliance validation*
- ✓ *2 Performance Query execution time, index usage*
- ✓ *3 Edge Cases Boundary conditions, NULL handling*



Real-World Business Value

FOR TOUR MANAGERS

- *15-25% cost reduction via route optimization*
- *Real-time P&L visibility per show*
- *Crew scheduling conflict prevention*
- *Data-driven venue selection*

FOR PROMOTERS

- *Artist ROI analysis for contracts*
- *Regional demand insights*
- *Venue performance comparison*

SYSTEM CAPABILITIES

- *Sub-second query response*
- *10x concurrent user capacity*
- *Zero overselling incidents*
- *Automated financial reconciliation*
- *100% data integrity (23/23 tests)*



FUTURE ENHANCEMENTS

- **REST API Development**

Enable integration with ticketing platforms and mobile apps

- **Machine Learning Integration**

- Demand forecasting for ticket pricing
 - Predictive route optimization
 - Automated crew recommendations

- **Mobile Application**

Crew check-in, real-time notifications, offline capability

Questions and Discussion

Thank You



Repository: github.com/SmithaJSU/jass-tour-management